

Gradual Liberalization and Resilient Trade Cooperation

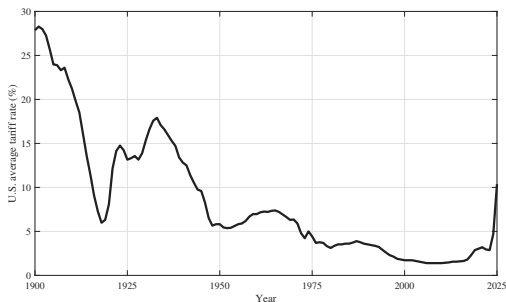
Marco Bassetto¹ Zhen Huo² José-Víctor Ríos-Rull³

¹Rice University

²Yale University

³University of Pennsylvania, UCL, CAERP, CEPR, NBER

Two Facts About Trade Policy



- **Liberalization is gradual:** RTAA agreements, successive GATT rounds, WTO commitments, multi-year phase-in schedules
- **Setbacks are temporary:** reversals need not be permanent

Standard Equilibrium Concepts Struggle

	Gradual liberalization?	Recovers from temporary reversal?
Markov (Staiger–Tabellini 1987)	no: stays high	–
Best SPE, grim trigger (Chari–Kehoe 1990)	maybe	no: permanent trade war
Organizational equilibrium	yes	yes

- Organizational equilibrium (Bassetto, Huo, Ríos-Rull 2024): a selection of subgame-perfect equilibria
 - cooperation among successive policymakers builds *slowly*
 - a disturbance delays liberalization but does not destroy it

This Paper

- Quantitative dynamic model of tariff policy *without commitment*
- Tariffs trade off
 - **now**: redistribution toward low-consumption workers, lower efficiency
 - **later**: lower return to capital, saving, and growth
- Separability $\Rightarrow$ dynamic problem simple to compute with exact hat algebra
- Results
 - Markov: tariff $\approx 12\%$ forever
 - Best sustainable (Markov reversion): complete liberalization, but one tremble $\Rightarrow$ permanent return to Markov
 - Organizational: gradual decline toward $\tau^B \approx 1.3\%$; after a temporary shock, liberalization simply resumes

Plan of the Talk

1. Organizational equilibrium: a primer
2. The trade model and its competitive equilibrium
3. Strategic government: Markov, best sustainable, organizational
4. Quantification
5. Results: tariff paths and resilience to shocks
6. Future directions

Time Inconsistency: Two Benchmarks

- Markov equilibrium
 - interesting comparative statics, outcome pinned down by fundamentals
 - ... but can be largely improved upon
- Sequential equilibrium / sustainable plans
 - can attain very good outcomes (folk theorem)
 - ... and very bad ones (folk theorem again)
 - relies on *self-punishment* as a threat
 - weak predictions: large set of equilibria
- Our view
 - good institutions and norms do not evolve overnight
 - collaboration across cohorts of decision makers builds slowly
 - ... and probably erodes slowly
 - look for an equilibrium concept that captures this

A Simple Example: Quasi-Geometric Discounting

- Preferences of the period- t decision maker

$$\Psi_t = \log c_t + \delta \sum_{\tau=1}^{\infty} \beta^{\tau} \log c_{t+\tau}, \quad \delta < 1 \Rightarrow \text{present bias}$$

- Technology: $k_{t+1} = k_t^{\alpha} - c_t$
- Action: saving rate $s_t \equiv k_{t+1}/k_t^{\alpha} \in [0, 1]$
 - always feasible, whatever k_t
 - given k_0 : $k_t = k_0^{\alpha^t} \prod_{j=0}^{t-1} s_j^{\alpha^{t-j-1}}$

Separability

- Lifetime utility of player t splits into state and actions

$$\underbrace{U(k_t, s_t, s_{t+1}, \dots)}_{\text{total payoff}} = \underbrace{\frac{\alpha(1 - \alpha\beta + \delta\alpha\beta)}{1 - \alpha\beta} \log k_t}_{\text{capital payoff}} + \underbrace{V(s_t, s_{t+1}, \dots)}_{\text{action payoff}}$$

$$\begin{aligned} V(s_t, \dots) &= \log(1 - s_t) + \frac{\delta\alpha\beta}{1 - \alpha\beta} \log s_t \\ &\quad + \delta \sum_{j \geq 1} \beta^j \left[\log(1 - s_{t+j}) + \frac{\alpha\beta}{1 - \alpha\beta} \log s_{t+j} \right] \end{aligned}$$

- Implications
 - preferences over action sequences are independent of k_t
 - the set of feasible sequences is independent of k_t
 - $\Rightarrow$ can compare payoffs across players who inherit different states

Benchmarks in the Example

- Markov (limit of finite horizon): constant saving rate

$$s^M = \frac{\alpha\beta\delta}{1 - \alpha\beta + \alpha\beta\delta}$$

- Ramsey (period-0 commitment)

$$s_0 = s^M, \quad s_t = s^R = \alpha\beta \quad t > 0$$

- Best *constant* saving rate

$$s^B = \frac{\alpha\beta\delta}{(1 - \beta + \delta\beta)(1 - \alpha\beta) + \delta\alpha\beta}, \quad s^M < s^B < s^R$$

Organizational Equilibrium: Characterization

Proposition (BHR 2024)

A sequence $\{\bar{s}_t\}_{t=0}^{\infty}$ is an organizational equilibrium if

1. **No restarting:** $V(\bar{s}_t, \bar{s}_{t+1}, \dots) = \bar{V} \quad \forall t \geq 0$
2. **Optimality:** no other sequence satisfying (1) attains a higher constant $\bar{V}$
3. **No delay:** $V(\bar{s}_0, \bar{s}_1, \dots) \geq \max_s V(s, \bar{s}_0, \bar{s}_1, \dots)$
 - Formally defined as the outcome of a subgame-perfect equilibrium satisfying state independence + these refinements
 - The proposition lets us work with *paths only*, like a competitive equilibrium

Where Do These Properties Come From?

- **No restarting** (looks forward)
 - akin to symmetry in reconsideration-proofness (Kocherlakota 1996)
 - if the equilibrium is too generous to player 0, player 1 wants to “become player 0”
 - rules out self-punishment and a privileged first player
- **Optimality**: no waste
- **No delay** (looks backward)
 - who starts the game? Time 0 could be any time
 - player 0 must not prefer to sit out and let the next player start the plan
 - “this time is different”: governments breaking with the past
 - in a game where history can be hidden, it is a *requirement* of sequential equilibrium game

Two Benchmarks Fail

- **Ramsey** $\{s^M, s^R, s^R, \dots\}$:

- no restarting fails

$$V(s^M, s^R, s^R, \dots) > V(s^R, s^R, s^R, \dots)$$

- player 1 prefers *copying* the proposal to *following* it
- same for the best SPE supported by Markov reversion

- **Markov**:

- no-restarting and no-delay hold

$$V(s^M, s^M, s^M, \dots)$$

- ... but we can do better

Best constant action

- $\{s^B, s^B, \dots\}$: no restarting and optimality OK, no delay fails

$$V(s^M, s^B, s^B, \dots) > V(s^B, s^B, s^B, \dots)$$

- player 0 plays Markov and lets the next player start the good plan
- But a path that *converges* to s^B can be implemented

Constructing the Organizational Equilibrium

- Constant action payoff

$V(s_t, s_{t+1}, \dots) = V(s_{t+1}, s_{t+2}, \dots) = \bar{V}$ gives a difference equation

$$\beta(1 - \delta) \log(1 - s_{t+1}) = \frac{\delta\alpha\beta}{1-\alpha\beta} \log s_t + \log(1 - s_t) - (1 - \beta)\bar{V}$$

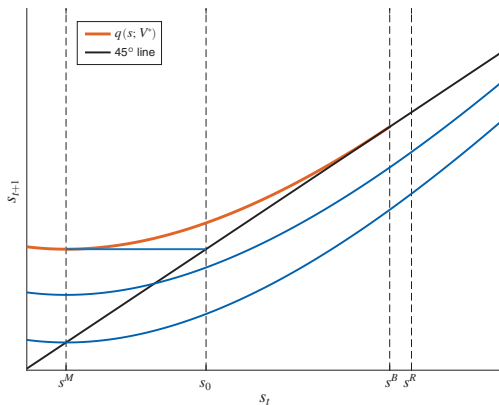
$$\iff s_{t+1} = q(s_t; \bar{V})$$

- Optimality: highest $\bar{V}$ with a well-behaved solution

$$\bar{V} = V^* \equiv V(s^B, s^B, \dots) \quad (q \text{ tangent to the } 45^\circ \text{ line at } s^B)$$

- No delay: $V(s^M, s_0, s_1, \dots) \leq V^*$
 - select $s_0 = q(s^M; V^*)$: no delay binds, most capital for successors

The Proposal Function



- Blue: lower $\bar{V}$; red: V^* , tangent at s^B
- Slope 0 at s^M , slope 1 at $s^B \Rightarrow$ convergence slows down near s^B

How to Compute

1. compute the best constant action a^B and its value V^*
2. find the recursive map of paths that deliver V^* every period
3. pick the initial action from the no-delay condition

One More Feature We Need

- **Large player + competitive agents**
 - government chooses τ_t ; private sector responds with s_t
 - if each $\{\tau_t\}$ implies a unique competitive $\{s_t(\tau)\}$ independent of the initial state, define government preferences over $\{\tau_t\}$ alone

Current Paper

- Application to foreign trade
 - Time inconsistency: protection good in the short run (favorable redistribution)
 - Protection bad in the long run (less capital, less growth)
- Show that the model can be used for quantitative work

Environment: Overview

- N countries (n, m), J sectors (i, j), discrete time
- Final use and intermediates: Cobb–Douglas across sectors, Armington (CES, elasticity ε) across origins
- Fixed mass L_{nj} of workers in each country-sector, inelastic supply
 - workers live hand to mouth, receive equal per-worker tariff rebate
- Capital: freely mobile across sectors and countries
 - owned by a representative world capitalist
 - aggregate country capital is an externality in production $\Rightarrow Ak$ growth

Production

- Country n 's final good (consumption and investment)

$$F_{n,t} = \prod_{j=1}^J \left(\frac{1}{\Phi_{nj}} \left[\sum_{m=1}^N (\Omega_{mnj}^c)^{1/\varepsilon} F_{mnj,t}^{(\varepsilon-1)/\varepsilon} \right]^{\varepsilon/(\varepsilon-1)} \right)^{\Phi_{nj}}$$

- Sectoral production:

$$y_{nj,t} = A_{nj} K_{n,t}^{\alpha_{k,j}} \ell_{nj,t}^{\alpha_{\ell,j}} x_{nj,t}^{\alpha_{x,j}}, \quad \alpha_{k,j} + \alpha_{\ell,j} + \alpha_{x,j} = 1$$

- $K_{n,t}$: aggregate capital in n , taken as given by firms
- Intermediate bundle $x_{nj,t}$:
 - same Cobb–Douglas/Armington nest as final good
 - elasticity ε
 - sector shares Ξ_{nji} , country/sector weights Ω_{mnji}^x

Capitalist consumption and investment

- World capitalist buys a Cobb–Douglas composite of country final goods:

$$\prod_{n=1}^N \left(\frac{Z_t^K}{\mu_n} \right)^{\mu_n}$$

- Z_t^K can be consumed or invested
- Capital depreciates at rate δ

Preferences

- Workers:

$$U_{nj} := \sum_{t=0}^{\infty} \beta^t \log C_{nj,t}$$

- Capitalists:

$$\sum_t \beta^t \frac{(C_t^K)^{1-\sigma}}{1-\sigma}$$

- Need $\sigma < 1$: saving rate increasing in the rate of return on capital
- (U.S.) government: utilitarian wrt U.S. workers

$$\sum_{j=1}^J L_{1j} U_{1j}$$

Policy Experiment

- Current Experiment:
 - Single strategic player, the United States
 - Chooses general tariff τ_t
 - Other countries mechanically retaliate with the same τ_t
- Possible Generalizations:
 - Cooperative solution across governments
 - Note: conflict analyzed is **over time**, not so much **across countries**
 - Any other unidimensional policy
 - Multidimensional policies potentially OK

Workers and Tariff Revenue

- Workers hand to mouth:

$$C_{nj,t} = \frac{W_{nj,t} + T_{n,t}/L_n}{P_{n,t}},$$

price indices, shares

- Revenue from final and intermediate imports,

$$T_{n,t} = T_{n,t}^c + T_{n,t}^x$$

$$T_{n,t}^c = P_{n,t} F_{n,t} \sum_j \Phi_{nj} \sum_{m \neq n} \frac{\tau_t}{1+\tau_t} \pi_{mnj,t}^c$$

$$T_{n,t}^x = \sum_j \alpha_{x,j} e_{nj,t} \sum_i \Xi_{nji} \sum_{m \neq n} \frac{\tau_t}{1+\tau_t} \pi_{mnij,t}^x$$

- Equal rebate $\Rightarrow$ larger proportional gain for low-wage sectors

Static Competitive Equilibrium

Definition

Given τ_t and K_t^W : allocations and prices $\{p_{nj,t}, W_{nj,t}, R_t\}$ such that

1. firms maximize profits (taking $K_{n,t}$ as given) and minimize input costs (factor shares)
2. final users minimize the cost of $F_{n,t}$ (price indices, shares)
3. workers consume $(W_{nj,t} + T_{n,t}/L_n)/P_{n,t}$
4. capitalists spend $R_t K_t^W$, a share μ_n in country n (capitalists)
5. variety, final-good, capital and labor markets clear
6. externality consistent: $K_{n,t} = \sum_j K_{nj,t}$
7. normalization: $\sum_{n,j} e_{nj,t} = 1$

Static Separability

Proposition (Static separability)

For any τ , the static equilibrium has a normalized representation independent of K^W : producer revenues, expenditure shares, tariff revenues, normalized prices $\hat{p} = K^W p$, and capital shares depend on τ but not on K^W .

- Revenues solve a linear system in trade shares only, with tariff wedge $\mathcal{D}_{mn,t} \equiv 1 + \tau_t \mathbf{1}_{m \neq n}$

$$e_{mi,t} = \sum_n \frac{\pi_{mni,t}^c}{\mathcal{D}_{mn,t}} \Phi_{ni} P_{n,t} F_{n,t} + \sum_{n,j} \frac{\pi_{mnij,t}^x}{\mathcal{D}_{mn,t}} \Xi_{nji} \alpha_{x,j} e_{nj,t}$$

- Unit cost: K^W cancels
 - scale K^W by a : quantities $\times a$, prices $\times 1/a$, revenues unchanged

Characterization of consumption and saving

- Worker consumption is proportional to world capital

$$C_{nj,t} = K_t^W \psi_{nj}(\tau_t), \quad \psi_{nj}(\tau) \equiv \frac{1}{\hat{P}_n} \left(\frac{\alpha_{\ell,j} e_{nj}}{L_{nj}} + \frac{T_n}{L_n} \right)$$

- Gross real return depends only on the tariff

$$G(\tau_t) \equiv 1 + \frac{\kappa_t}{\hat{P}_t^K} - \delta, \quad \kappa_t = \sum_{n,j} \alpha_{k,j} e_{nj,t}$$

- Dynamics given $\{\tau_t\}$

$$K_{t+1}^W = s_t G(\tau_t) K_t^W, \quad \left(\frac{1-s_t}{s_t} \right)^{-\sigma} = \beta G(\tau_{t+1})^{1-\sigma} (1-s_{t+1})^{-\sigma}$$

- Competitive equilibrium: each tariff sequence $\Rightarrow$ unique $\{s_t\}$, independent of K_0^W

Policymaker's Objective

- Utilitarian over workers of the lead country h (U.S.)

$$\frac{1}{L_1} \sum_j L_{1j} \log C_{1j,t} = \log K_t^W + \chi(\tau_t)$$

- Iterating on $K_{t+1}^W = s_t G(\tau_t) K_t^W$:

$$U_t = \frac{1}{1-\beta} \log K_t^W + \sum_{k \geq 0} \beta^k H(\tau_{t+k}, s_{t+k})$$

$$H(\tau, s) \equiv \underbrace{\chi(\tau)}_{\text{static, workers}} + \frac{\beta}{1-\beta} \underbrace{[\log s + \log G(\tau)]}_{\text{accumulation}}$$

- Separable: state payoff + action payoff

$$\mathcal{V}_t = \sum_k \beta^k H(\tau_{t+k}, s_{t+k})$$

Where the Time Inconsistency Comes From

- s_t depends on $\tau_{t+1}, \tau_{t+2}, \dots$ but not on τ_t
- With commitment at $t = 0$
 - τ_0 has no effect on saving: free to protect
 - $\tau_1, \tau_2, \dots$ depress $s_0, s_1, \dots$: keep them low
- Reoptimizing at $t = 1$: s_0 is sunk, so τ_1 looks like τ_0 did
- Same structure as quasi-geometric discounting, with
 - action = tariff, chosen by the government
 - private response = capitalists' saving

Markov Equilibrium

- Policymaker chooses τ_t taking continuation tariffs and saving as given
- Current s_t is independent of τ_t , so

$$\tau^M \in \arg \max_{\tau \geq 0} \left\{ \chi(\tau) + \frac{\beta}{1-\beta} \log G(\tau) \right\}$$

- internalizes the resource effect of τ_t on capitalists' wealth
 - ignores the effect of expected tariffs on the *saving rate*
- Stationary: $\tau_t^M = \tau^M$, with

$$(s^M)^\sigma = \beta G(\tau^M)^{1-\sigma}, \quad \mathcal{V}^M = \frac{H(\tau^M, s^M)}{1-\beta}$$

Best Sustainable Equilibrium (Markov Reversion)

- Follow $\{\tau_t^S\}$; after any deviation, τ^M forever
- Sustainability: $\mathcal{V}_t^S \equiv \sum_k \beta^k H(\tau_{t+k}^S, s_{t+k}^S) \geq \mathcal{V}^M \quad \forall t$
- Best sustainable: $\max \mathcal{V}_0$ s.t. Euler equations and sustainability
- Features
 - period 0 is special: $\mathcal{V}_0 > \mathcal{V}_t$; if constraints slack $\Rightarrow$ Ramsey
 - punishment is *self-inflicted*: future players impose a policy they dislike to discipline past players
 - vulnerable to renegotiation / reconsideration

Organizational Equilibrium: Best Constant Tariff

- Step 1: best constant policy. With τ constant,
 $s(\tau)^\sigma = \beta G(\tau)^{1-\sigma}$, so

$$\tau^B \in \arg \max_{\tau \geq 0} \left\{ \chi(\tau) + \frac{\beta}{\sigma(1-\beta)} \log G(\tau) \right\}$$

- Compare with Markov: weight on $\log G$ is $1/\sigma$ times larger
- Define $s^B \equiv s(\tau^B)$, $H^B \equiv H(\tau^B, s^B)$, $\mathcal{V}^B = H^B/(1-\beta)$

Step 2: Recursive Representation

- Constant action payoff: $\mathcal{V}_t^O = \mathcal{V}^B$ for all $t \iff$

$$H(\tau_t^O, s_t^O) = H^B \implies s_t^O = s^O(\tau_t^O)$$

- Combine with the capitalists' Euler equation

$$\left[\frac{1 - s^O(\tau_t)}{s^O(\tau_t)} \right]^{-\sigma} = \beta G(\tau_{t+1})^{1-\sigma} [1 - s^O(\tau_{t+1})]^{-\sigma}$$

- Implicitly defines the transition map

$$\boxed{\tau_{t+1} = \mathcal{T}(\tau_t)}, \quad \mathcal{T}(\tau^B) = \tau^B, \quad \tau^B < \mathcal{T}(\tau) < \tau \text{ for } \tau > \tau^B$$

- Mapping implies gradualism:
 - Lower tariff today rewarded by further progress tomorrow
 - Steady state reached asymptotically

Step 3: No Delay Pins Down the Start

- Alternative for player 0: play τ^M today, let the plan start tomorrow
 - saving would be s_{-1}^O , from the Euler eq. with $\tau^M, \tau_0^O, \tau_1^O, \dots$
- No delay: $H(\tau_0^O, s_0^O) \geq H(\tau^M, s_{-1}^O)$
- Lowest tariff satisfying it

$$\tau_0^O = \mathcal{T}(\tau^M)$$

- then $s_{-1}^O = s^O(\tau^M)$ and $H(\tau^M, s^O(\tau^M)) = H^B$: player 0 is indifferent
- Time 0 is *not* special: τ_0^O is on the continuation of any history in which someone played τ^M

Resilience to Trembles

- Suppose an unexpected event sets today's tariff to $\tilde{\tau}$
- **Organizational**: continue with $\mathcal{T}(\tilde{\tau}), \mathcal{T}(\mathcal{T}(\tilde{\tau})), \dots$
 - the player choosing $\tilde{\tau}$ gets exactly H^B : sufficient deterrent
 - no permanent loss for anyone
 - tariffs decline monotonically back to τ^B
 - a shock resets the *starting point*, not the *destination*
- Contrast with **Best sustainable**:
 - $\tau_t = \tau^M$ for all t after the shock
 - even when a milder threat would suffice, would need a discrete drop in utility from tomorrow

Comparing the Equilibria

	Markov	Best sustainable	Organizational
On-path tariff	τ^M forever	front-loaded, faster decline	slower decline to τ^B
Period 0	not special	special: $\mathcal{V}_0 > \mathcal{V}_t$	not special
After a tremble	τ^M	τ^M forever	restart from $\tilde{\tau}$
Punishment	none	permanent, self-inflicted	proportional

- Trade-off: faster liberalization when nothing goes wrong vs. robustness when policy occasionally departs from plan

Exact-Hat Representation

Proposition (Sufficient statistics)

Conditional on baseline final and intermediate expenditure shares, production shares $(\alpha_k, \alpha_\ell, \alpha_x)$, capitalist weights $\{\mu_n\}$ and ε , tariff counterfactuals do not require $\{A_{nj}\}$, Armington weights $\{\Omega\}$, or labor endowments.

- Baseline shares summarize technology, tastes, frictions, endowments
- Only U.S. employment *shares* λ_{hj} are needed for $\chi(\tau)$
- Map each $\tau \mapsto (\chi(\tau), G(\tau))$; fast enough to solve the dynamic game repeatedly
- Caveat: the *common* technology level matters for growth
 - calibrated to 2% long-run growth in the organizational equilibrium

Data

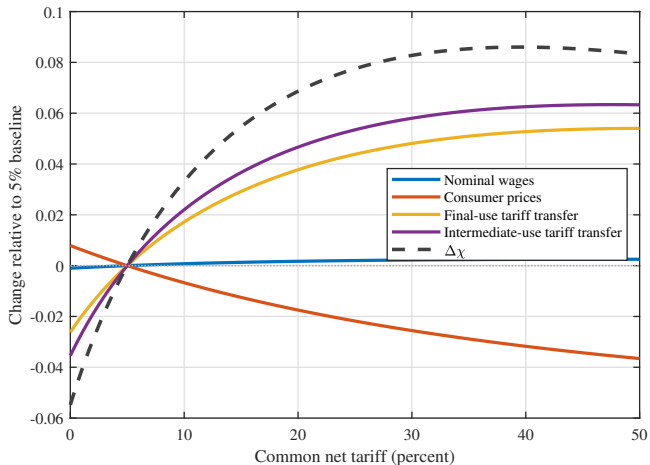
- World = G7
- 30 sectors (harmonization of Huo–Levchenko–Pandalai–Nayar 2025)
 - agriculture, manufacturing, utilities, construction, trade, transportation, business services, public and social services
- **WIOD** (2013)
 - bilateral final-use flows by destination and supplying country-sector
 - country-sector $\times$ country-sector intermediate flows
 - full domestic + international matrix disciplines the network
- **KLEMS** (2009 vintage)
 - gross output, value added, factor inputs $\Rightarrow$ sectoral factor shares
 - U.S. sectoral employment (2007) $\Rightarrow \lambda_{hj}$

Calibration

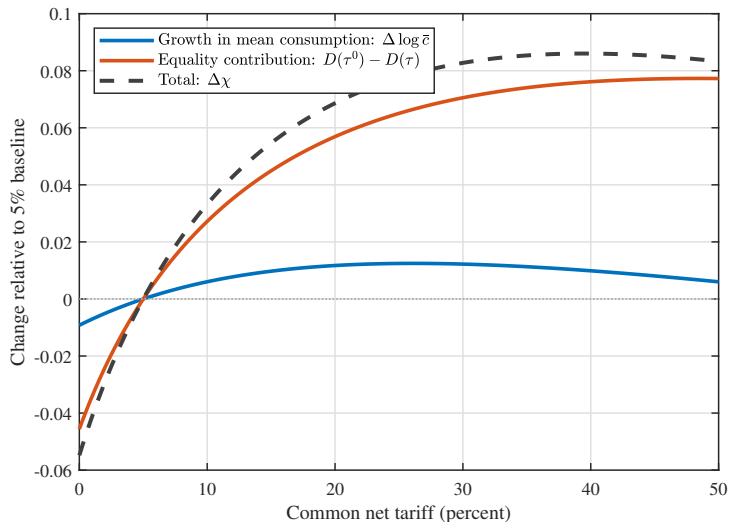
Param.	Value	Source	Role
ε	3.5	literature	Armington elasticity
β	0.96	standard	discount factor (annual)
δ	0.10	standard	depreciation
σ	0.5	chosen	capitalists' curvature (IES = 2)
A	0.43	2% growth	common technology level
$\alpha_{\ell,j}, \alpha_{k,j}, \alpha_{x,j}$	by sector	KLEMS	factor shares
$\Phi_{nj}, \bar{\pi}_{mnj}^c$	—	WIOD	final-use shares
$\Xi_{nji}, \bar{\pi}_{mnij}^x$	—	WIOD	intermediate shares
L_{1j}	—	KLEMS	U.S. employment

- Baseline common tariff set at 5% to convert hat changes into levels

Tariffs and U.S. Worker Welfare: One Shot

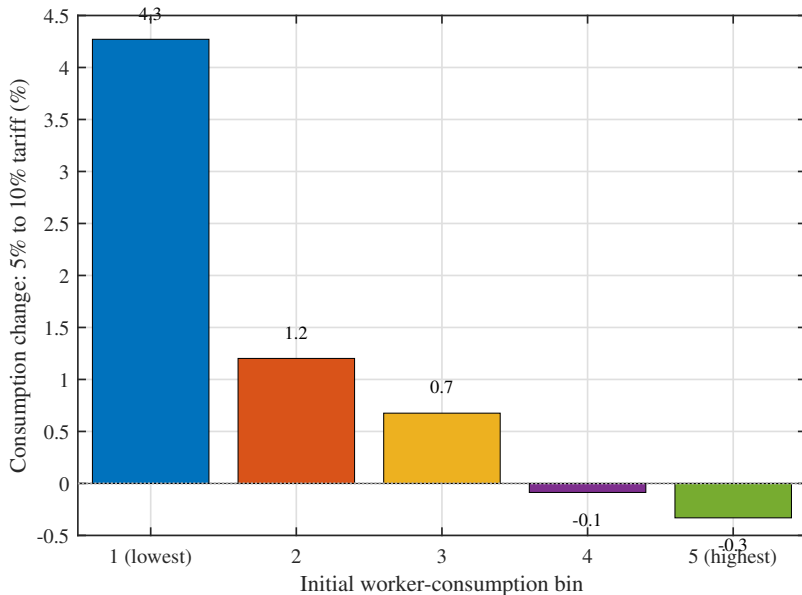


It Is About Equality, Not Average Consumption

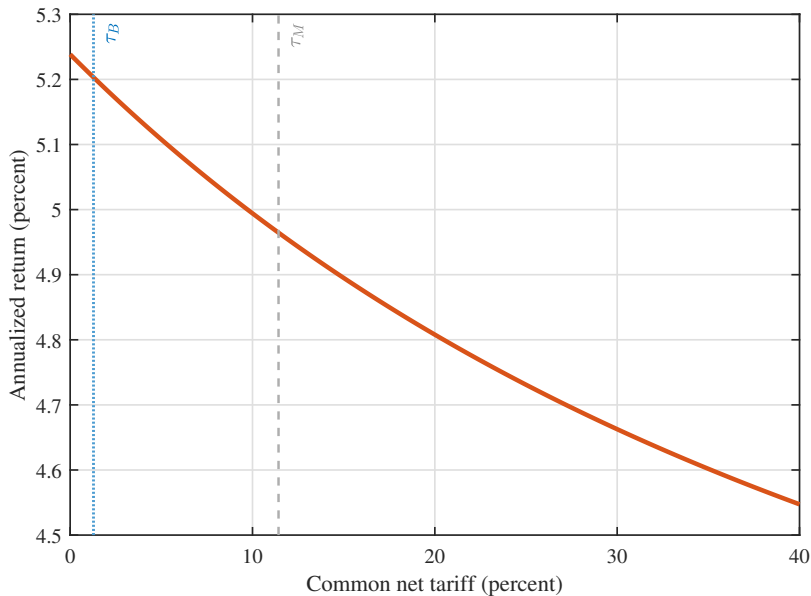


mean vs. equality contribution

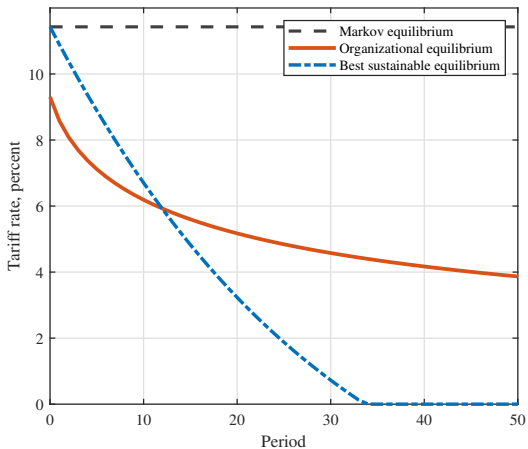
Distribution Effects Across Sectors



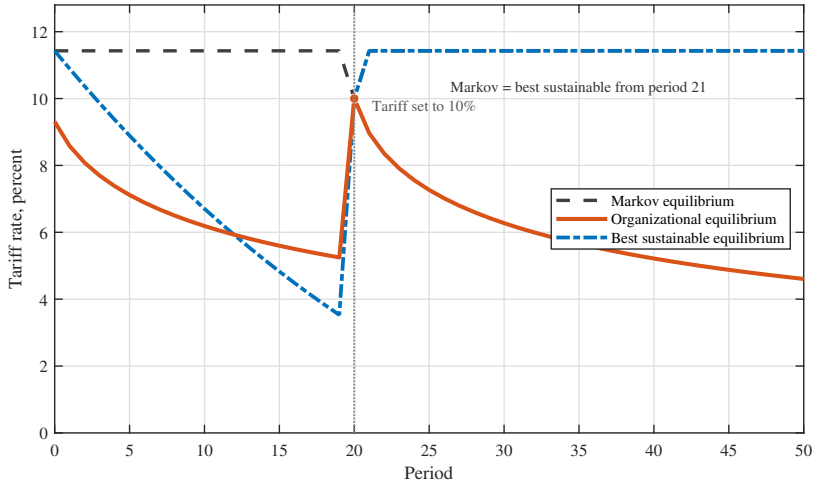
The Cost: The Return to Capital



Equilibrium Tariff Paths



A Temporary Protectionist Shock



Conclusions and Future Directions

- Quantitative theory of gradual liberalization, resilient to temporary protection
 - Markov: high protection forever
 - Best sustainable: fast, but fragile to a single tremble
 - Organizational: gradual, and returns to the same low-tariff destination after a shock

Future Directions

- multiple strategic countries (requires extending the equilibrium notion)
- explicit uncertainty; incomplete information

Backup: A Game Where Period 0 Is Not Special

- Past actions unobservable until an “organization” records play
- Opportunity to start record keeping arrives at a random, unobserved date $\hat{t}$ (full support on $\mathbb{N}$)
- Each player may continue the organization, start a new one (only its own action is recorded), or discontinue it
- Weak separability (multiplicative-additive form) $\Rightarrow$ state-independent sequential equilibria
- Result: path from $\hat{t}$ on, in any state-independent sequential eq. satisfying no restarting and optimality, is an organizational equilibrium
- No delay is not an extra refinement: otherwise the first player able to keep records would “pass the buck” and let the next one believe it is first

Backup: Price Indices and Expenditure Shares

- Delivered price: $p_{mnj,t}^d = \mathcal{D}_{mn,t} p_{mj,t}$, wedge $\mathcal{D}_{mn,t} \equiv 1 + \tau_t \mathbf{1}_{m \neq n}$
- Final use: price indices and origin shares

$$Q_{nj,t}^c = \left[\sum_m \Omega_{mnj}^c (p_{mnj,t}^d)^{1-\varepsilon} \right]^{\frac{1}{1-\varepsilon}}, \quad P_{n,t} = \prod_j (Q_{nj,t}^c)^{\Phi_{nj}}$$

$$\pi_{mnj,t}^c = \frac{\Omega_{mnj}^c (p_{mnj,t}^d)^{1-\varepsilon}}{\sum_r \Omega_{rnj}^c (p_{rnj,t}^d)^{1-\varepsilon}}$$

- share of n 's final spending on (m, j) : $\Phi_{nj} \pi_{mnj,t}^c$
- Intermediates: same nest, weights Ω^x , sector shares Ξ_{nji}

$$P_{nj,t}^x = \prod_i (Q_{nji,t}^x)^{\Xi_{nji}}$$

- $\pi_{mnij,t}^x$: share of origin m in (n, j) 's purchases of sector- i inputs
- Other baseline frictions (non-tariff barriers, shipping) live in Ω

Backup: Firms and Factor Shares

- Sector j in country n : competitive firms, common rental rate R_t
- First-order conditions

$$R_t k_{nj,t} = \alpha_{k,j} e_{nj,t}, \quad W_{nj,t} L_{nj} = \alpha_{\ell,j} e_{nj,t}, \quad P_{nj,t}^x x_{nj,t} = \alpha_{x,j} e_{nj,t}$$

where $e_{nj,t} \equiv p_{nj,t} y_{nj,t}$ is producer revenue

- Summing over (n,j) : payments to capital

$$\kappa_t \equiv R_t K_t^W = \sum_{n,j} \alpha_{k,j} e_{nj,t}, \quad \frac{k_{nj,t}}{K_t^W} = \frac{\alpha_{k,j} e_{nj,t}}{\kappa_t}$$

Backup: Capitalists

- Composite price $P_t^K = \prod_n P_{n,t}^{\mu_n}$; share μ_n spent in country n
- Problem

$$\max \sum_t \beta^t \frac{(C_t^K)^{1-\sigma}}{1-\sigma} \quad \text{s.t.} \quad C_t^K + K_{t+1}^W = \left(1 + \frac{R_t}{P_t^K} - \delta\right) K_t^W$$

- Saving rate $K_{t+1}^W = s_t \left(1 + \frac{R_t}{P_t^K} - \delta\right) K_t^W$ satisfies

$$\left(\frac{1-s_t}{s_t}\right)^{-\sigma} = \beta \left(1 + \frac{R_{t+1}}{P_{t+1}^K} - \delta\right)^{1-\sigma} (1-s_{t+1})^{-\sigma}$$

- s_t depends only on *future* returns, hence on *future* tariffs
- $\sigma < 1$: saving increasing in the return $\Rightarrow$ protection hurts accumulation

Backup: Proof of Static Separability

- Take an equilibrium with K^W ; for $a > 0$ set $\tilde{K}^W = aK^W$ and

$$\tilde{K}_{nj} = aK_{nj}, \quad \tilde{X}_{nj} = aX_{nj}, \quad \tilde{Y}_{nj} = aY_{nj}, \quad \tilde{F}_n = aF_n$$

$$\tilde{p}_{nj} = p_{nj}/a, \quad \tilde{R} = R/a, \quad \tilde{W}_{nj} = W_{nj}$$

- Feasibility uses $\gamma_j = \alpha_{\ell,j}$:

$$A_{nj} \tilde{K}_n^{\gamma_j} \tilde{K}_{nj}^{\alpha_{k,j}} L_{nj}^{\alpha_{\ell,j}} \tilde{X}_{nj}^{\alpha_{x,j}} = a^{\alpha_{\ell,j} + \alpha_{k,j} + \alpha_{x,j}} Y_{nj} = aY_{nj}$$

- All delivered prices divided by $a \Rightarrow$ Armington shares unchanged; CES/CD price indices divided by a
- Factor payments, revenues e_{nj} , κ , final expenditure, tariff revenue unchanged
- Normalized prices $K^W p$ and capital shares $K_{nj}/K^W = \alpha_{k,j} e_{nj}/\kappa$ unchanged

Backup: Exact-Hat System

With $\dot{z} = z'/z$, $d_{mn} = \mathcal{D}'_{mn}/\mathcal{D}_{mn}$:

$$\pi_{mnj}^{c'} = \frac{\pi_{mnj}^c (d_{mn} \dot{p}_{mj})^{1-\varepsilon}}{\sum_r \pi_{rnj}^c (d_{rn} \dot{p}_{rj})^{1-\varepsilon}}, \quad \dot{Q}_{nj}^c = \left[\sum_m \pi_{mnj}^c (d_{mn} \dot{p}_{mj})^{1-\varepsilon} \right]^{\frac{1}{1-\varepsilon}}$$

(same for intermediates), $\dot{P}_n = \prod_j (\dot{Q}_{nj}^c)^{\Phi_{nj}}$, $\dot{P}^K = \prod_n \dot{P}_n^{\mu_n}$

$$\dot{p}_{nj} = \dot{\kappa}^{\alpha_{k,j}} \dot{e}_{nj}^{\alpha_{\ell,j}} (\dot{P}_{nj}^x)^{\alpha_{x,j}} (\dot{v}_n^K)^{-\alpha_{\ell,j}}, \quad \vartheta_n^K = \frac{\sum_j \alpha_{k,j} e_{nj}}{\kappa}$$

$$E'_n = \frac{\sum_j \alpha_{\ell,j} e'_{nj} + T_n^{x'} + \mu_n \kappa'}{1 - g_n^{c'}}, \quad e'_{mi} = \sum_n \frac{\pi_{mni}^{c'}}{\mathcal{D}'_{mn}} \Phi_{ni} E'_n + \sum_{n,j} \frac{\pi_{mnij}^{x'}}{\mathcal{D}'_{mn}} \Xi_{nji} \alpha_{x,j} e'_{nj}, \quad \sum e' = 1$$

Outputs:

$$\frac{r'}{r} = \frac{\dot{\kappa}}{\dot{P}^K}, \quad \frac{C'_{hj}}{C_{hj}} = \frac{1}{\dot{P}_h} \frac{\alpha_{\ell,j} e'_{hj} + \lambda_{hj} T'_h}{\alpha_{\ell,j} e_{hj} + \lambda_{hj} T_h}$$

No A_{nj} , Ω , or L_{nj} levels appear.

Backup: Welfare Decompositions

- $\Delta\chi(\tau) = \mathcal{W}(\tau) + \mathcal{P}(\tau) + \mathcal{R}^c(\tau) + \mathcal{R}^x(\tau)$

$$\mathcal{W} = \sum_j \lambda_{hj} \log \frac{W_{hj}(\tau)}{W_{hj}(0)}, \quad \mathcal{P} = -\log \frac{P_h(\tau)}{P_h(0)}$$

- Rebate terms: two-factor Shapley split of

$$\mathcal{R}^c + \mathcal{R}^x = \sum_j \lambda_{hj} \log \frac{W_{hj} + T_h^c/L_h + T_h^x/L_h}{W_{hj}}$$

averaging over the two orders in which the revenue sources are added

- Mean vs. equality: $\bar{\psi}_h = \sum_j \lambda_{hj} \psi_{hj}$,
 $D = \log \bar{\psi}_h - \sum_j \lambda_{hj} \log \psi_{hj} \geq 0$

$$\Delta\chi = \log \frac{\bar{\psi}_h(\tau)}{\bar{\psi}_h(0)} + [D(0) - D(\tau)]$$